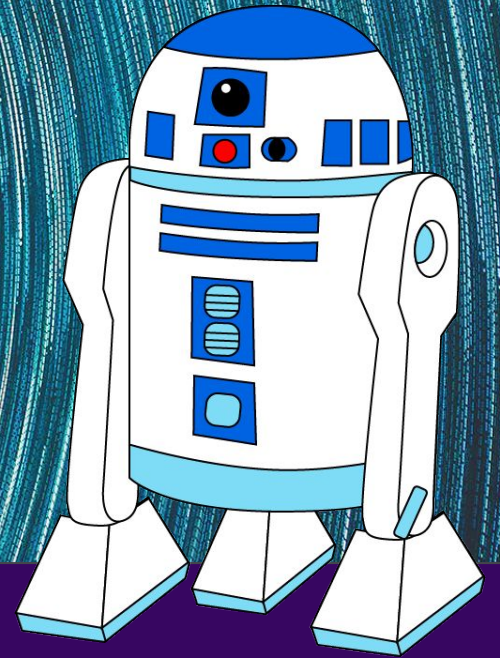


CIS 421/521:
ARTIFICIAL INTELLIGENCE

NLP: Vector Semantics

Jurafsky and Martin Chapter 6



Word Meaning

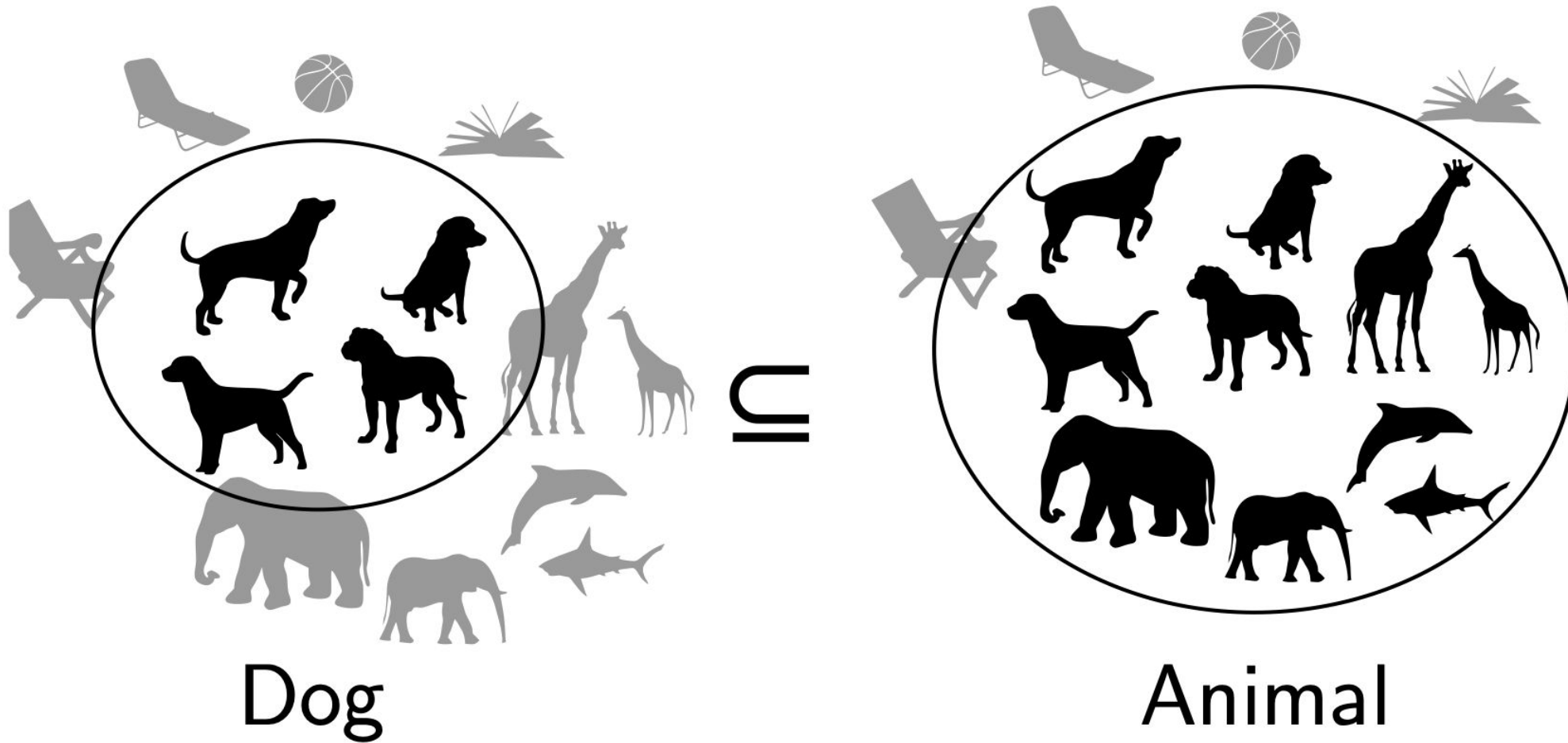
How should we **represent** the **meaning** of a word?

In N-gram LMs we represented words as a string of letters or as an index in a vocabulary list.

Ideally, we want a meaning representation to encode:

1. **Synonyms** – words that have similar meanings
2. **Antonyms** – words that have opposite meanings
3. **Connotations** – words that are positive or negative
4. **Semantic Roles** – *buy, sell, and pay* are different parts of the same underlying *purchasing* event
5. Support for **entailment**

Entailment in formal semantics



Entailment in formal semantics

All animals have an ulnar artery

$\Rightarrow$

All dogs have an ulnar artery

- + Mathematically well-understood
- + Powerful machinery for handling logical operations
- Knowledge must come from somewhere else

Noun

- S: (n) **dog**, domestic dog, Canis familiaris (a member of the genus *Canis* (probably descended from the common wolf) that has been domesticated by man since prehistoric times; occurs in many breeds) *"the dog barked all night"*
- S: (n) frump, **dog** (a dull unattractive unpleasant girl or woman) *"she got a reputation as a frump"; "she's a real dog"*
- S: (n) **dog** (informal term for a man) *"you lucky dog"*
- S: (n) cad, bounder, blackguard, **dog**, hound, heel (someone who is morally reprehensible) *"you dirty dog"*
- S: (n) frank, frankfurter, hotdog, hot dog, **dog**, wiener, wienerwurst, weenie (a smooth-textured sausage of minced beef or pork usually smoked; often served on a bread roll)
- S: (n) pawl, detent, click, **dog** (a hinged catch that fits into a notch of a ratchet to move a wheel forward or prevent it from moving backward)
- S: (n) andiron, firedog, **dog**, dog-iron (metal supports for logs in a fireplace) *"the andirons were too hot to touch"*

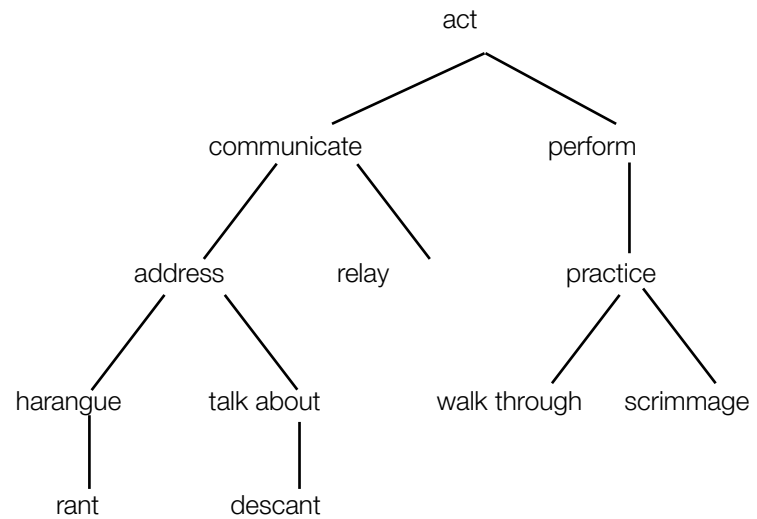
Verb

Noun

- S: (n) **dog**, domestic dog, Canis familiaris (a member of the genus Canis (probably descended from the common wolf) that has been domesticated by man since prehistoric times; occurs in many breeds) *"the dog barked all night"*
 - direct hyponym / full hyponym
 - part meronym
 - member holonym
 - direct hypernym / inherited hypernym / sister term
 - S: (n) canine, canid (any of various fissiped mammals with nonretractile claws and typically long muzzles)
 - S: (n) domestic animal, domesticated animal (any of various animals that have been tamed and made fit for a human environment)
- S: (n) frump, **dog** (a dull unattractive unpleasant girl or woman) *"she got a reputation as a frump"; "she's a real dog"*
- S: (n) **dog** (informal term for a man) *"you lucky dog"*
- S: (n) cad, bounder, blackguard, **dog**, hound, heel (someone who is morally reprehensible) *"you dirty dog"*
- S: (n) frank, frankfurter, hotdog, hot dog, **dog**, wiener, wienerwurst, weenie

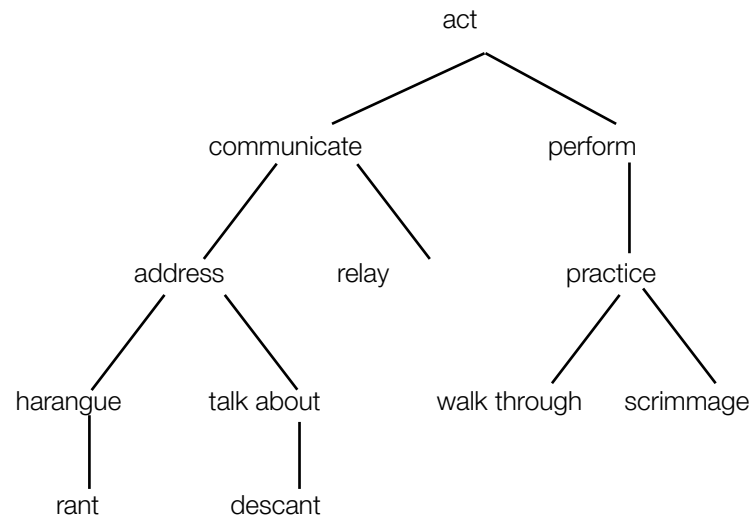
- S: (n) canine, canid (any of various fissiped mammals with nonretractile claws and typically long muzzles)
 - S: (n) carnivore (a terrestrial or aquatic flesh-eating mammal)
"terrestrial carnivores have four or five clawed digits on each limb"
 - S: (n) placental, placental mammal, eutherian, eutherian mammal (mammals having a placenta; all mammals except monotremes and marsupials)
 - S: (n) mammal, mammalian (any warm-blooded vertebrate having the skin more or less covered with hair; young are born alive except for the small subclass of monotremes and nourished with milk)
 - S: (n) vertebrate, craniate (animals having a bony or cartilaginous skeleton with a segmented spinal column and a large brain enclosed in a skull or cranium)
 - S: (n) chordate (any animal of the phylum Chordata having a notochord or spinal column)
 - S: (n) animal, animate being, beast, brute, creature, fauna (a living

Lexical Semantics

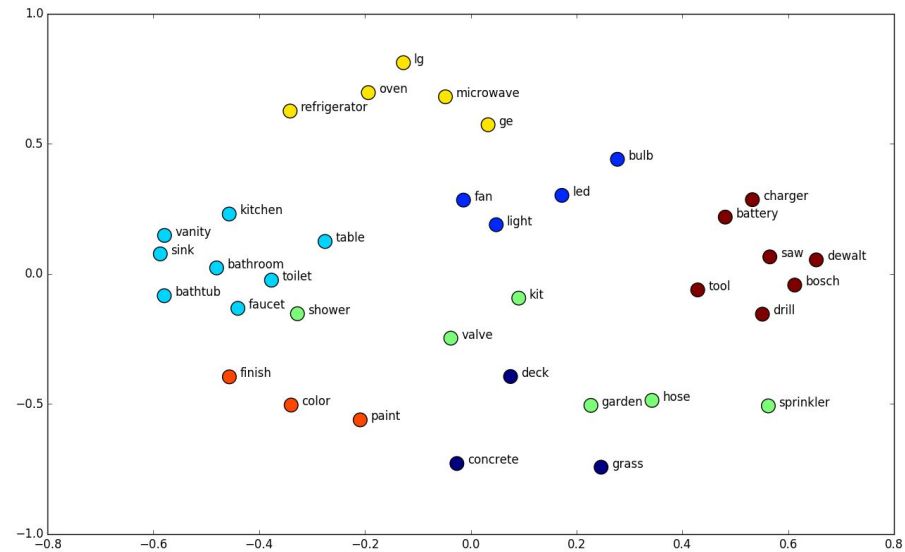


WordNet

Lexical Semantics

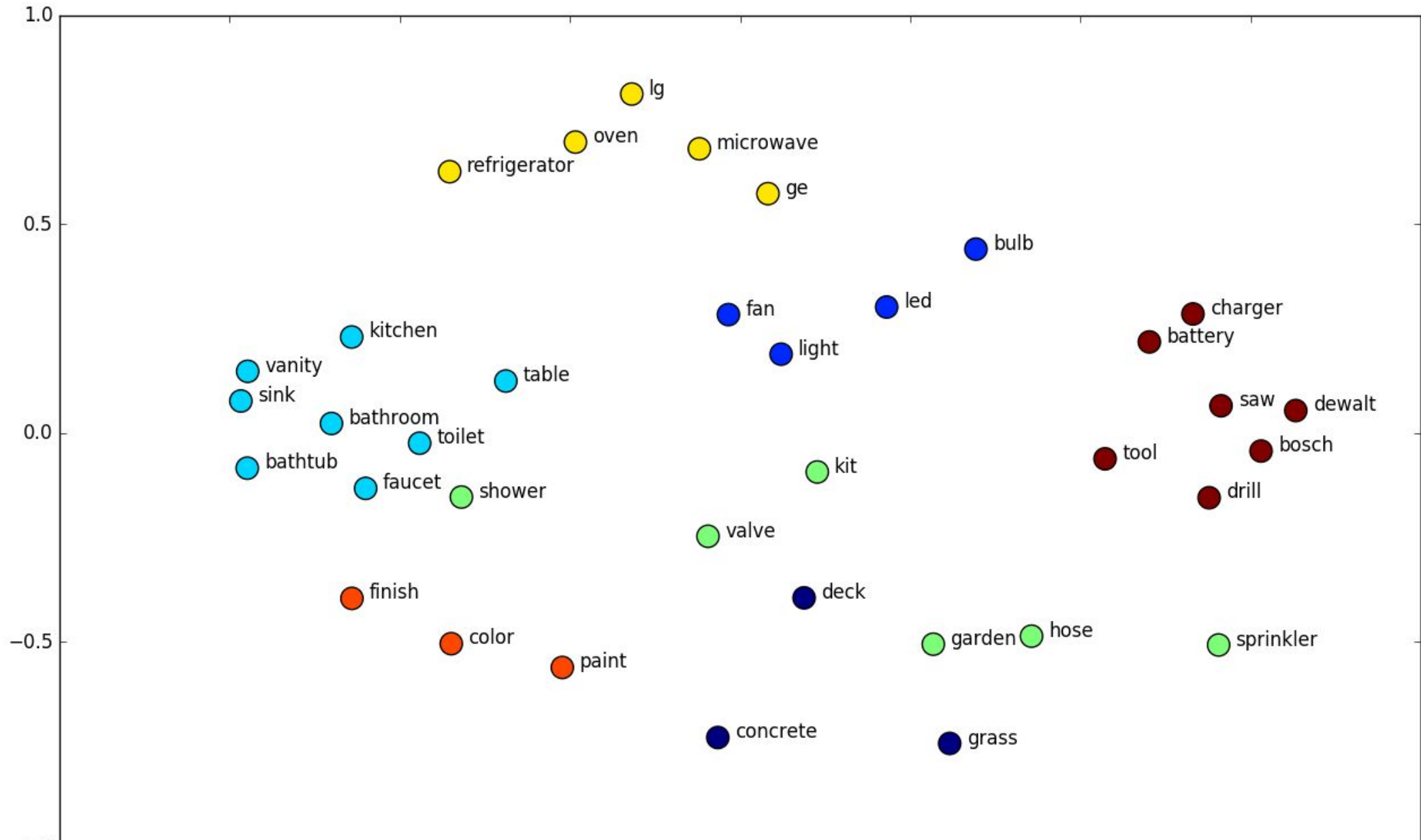


WordNet



Vector Space Models

Vector Space Models



Dictionary Definitions

Noun

1. A small insect.
2. A harmful microorganism, as a bacterium or virus.
3. An enthusiastic, almost obsessive, interest in something.
'they caught the sailing bug'
4. A miniature microphone, typically concealed in a room or telephone, used for surveillance.
5. An error in a computer program or system.

Verb

1. Conceal a miniature microphone in (a room or telephone) in order to monitor or record someone's conversations.
2. Annoy or bother (someone)

Polysemy

A word that has multiple meanings is called **polysemous**. We call each of these aspects of the meaning of *bug* a **word sense**.

Polysemy can make interpretation difficult.

What if someone types “caught a bug” into Google?

Word sense disambiguation is the task of determining which sense of a word is being used in a context.

Synonymy

When one word has a sense whose meaning is nearly identical to a sense of another word then those two words are **synonyms**.

glitch/error

microbe/bacterium

insect/pest

microphone/wire

Formally, two words are synonymous if they are **substitutable** one for the other in **any sentence** without changing the **truth conditions** of the sentence.

In logic, that means the two words carry the same **propositional meaning**.

Principle of Contrast

Linguists assume that **a difference in form** is always associated with a **difference in meaning**.

While substitutions like *water/H₂O* or *father/dad* are truth preserving, the words are still not identical in meaning.

H₂O is used in scientific contexts, but not general texts like hiking guides

Father is a more formal version of *dad*.

It is possible that no two words have **absolutely identical** meaning.

Word similarity

Most words don't have many **synonyms**, but they do have a lot of **similar** words. *Cat* is not a synonym of *dog*, but *cats* and *dogs* are certainly similar words.

“**fast**” is similar to “**rapid**”

“**tall**” is similar to “**height**”

Useful for applications like question answering

2:12



W

Mo
not

How tall is mount Everest

Tap to Edit >

**According to Wikipedia,
it's 29,029'.**



KNOWLEDGE

Mount Everest

Earth's highest mountain, part of the Himalaya between Nepal and China



Mount Everest, known in Nepali as Sagarmāthā and in Tibetan as Chomolungma, is Earth's highest mountain above sea level, located in the Mahalangur Himal sub-range of the Himalayas. The international border between China and Nepal runs across its summit point. The current official elevation of 8,848 m, recognised by China and Nepal, was established by a 1955 Indian survey an... [more](#)

Elevation above sea level 29,028 ft

Named after George Everest >



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Mount Everest

From Wikipedia, the free encyclopedia

Coordinates: 27°59′17″N 86°55′31″E﻿ / ﻿

"Everest" redirects here. For other uses, see *Everest (disambiguation)*.



This article's **tone or style may not reflect the encyclopedic tone used on Wikipedia**. See Wikipedia's [guide to writing better articles](#) for suggestions. (October 2017) ([Learn how and when to remove this template message](#))

Mount Everest, known in **Nepali** as **Sagarmāthā** and in **Tibetan** as **Chomolungma**, is **Earth's highest mountain** above **sea level**, located in the **Mahalangur Himal** sub-range of the **Himalayas**. The international border between **China** (**Tibet Autonomous Region**) and **Nepal** (**Province No. 1**) runs across its summit point

The current of recognised by 1955 Indian si Chinese surve the rock height m. There follo Nepal as to whether the official height should be the rock height (8,844 m., China) or the snow height (8,848 m., Nepal). In 2010, an agreement was reached by both sides that the height of Everest is 8,848 m, and Nepal recognises China's claim that the rock height of Everest is 8,844 m.^[5]

In 1865, Everest was given its official English name by the **Royal Geographical Society**, upon a recommendation by **Andrew Waugh**, the British **Surveyor General of India**. As there appeared to be several different local names, Waugh chose to name the

Mount Everest

सागरमाथा (Sagarmāthā)
ཇོ་མོ་གླང་མ (Chomolungma)
珠穆朗玛峰 (Zhūmùlǎngmǎ Fēng)

height (8,844 m., China) or the snow height (8,848 m., Nepal). In 2010, an agreement was reached by both sides that the height of Everest is 8,848 m, and Nepal recognises China's claim that the rock height of Everest is 8,844 m.^[5]

Everest's north face from the Tibetan plateau

Highest point

Elevation 8,848 metres (29,029 ft)^[1]
Ranked 1st

Prominence Ranked 1st
(Notice special definition for Everest)

Listing Seven Summits
Eight-thousander
Country high point
Ultra

Coordinates 27°59′17″N 86°55′31″E﻿ / ﻿^[2]

Geography

Word similarity

Can similar words be substituted in any sentence without changing its truth conditions?
No.

How can we measure whether words are similar? One way is to ask humans to judge how similar one word is to another.

Word 1	Word 2	Similarity Score
Vanish	Disappear	9.8
Tiger	Cat	7.4
Love	Sex	6.8
Muscle	Bone	3.6
Cucumber	Professor	0.3

Word Relatedness

Words can still be **related** in ways other than being similar to each other.

Coffee and *Cup* are **not similar** because they don't share any features

1. *coffee* is a plant or a beverage,
2. *cup* is a manufactured object made in a useful shape

But they're **related** by co-participating in the same **event**.

Relatedness is measured with **word association** tests in psychology.

A **semantic field** is a set of words which cover a semantic domain and bear structured relations with each other.

Hospitals: *surgeon, scalpel, nurse, anesthetic, hospital*

Restaurants: *waiter, menu, plate, food, chef*

Houses: *family, door, roof, kitchen, bed*

Semantic Roles

An **event** like a commercial transaction described with different **verbs**

1. *buy* (the event from the perspective of the buyer),
2. *sell* (from the perspective of the seller),
3. *pay* (focusing on the monetary aspect),

Or with nouns like *buyer*.

Frames encode **semantic roles** (like *buyer, seller, goods, money*), and the words in a sentence that take on these roles.

Connotation

Words have **affective meanings** or **connotations**. Three important dimensions of affective meaning.

1. *Valence* – the pleasantness of the stimulus
2. *Arousal* – the intensity of emotion provoked by the stimulus
3. *Dominance* – the degree of control exerted by the stimulus

	Valence	Arousal	Dominance
courageous	8.05	5.5	7.38
music	7.67	5.57	6.5
heartbreak	2.45	5.65	3.58
cub	6.71	3.95	4.24
life	6.68	5.59	5.89

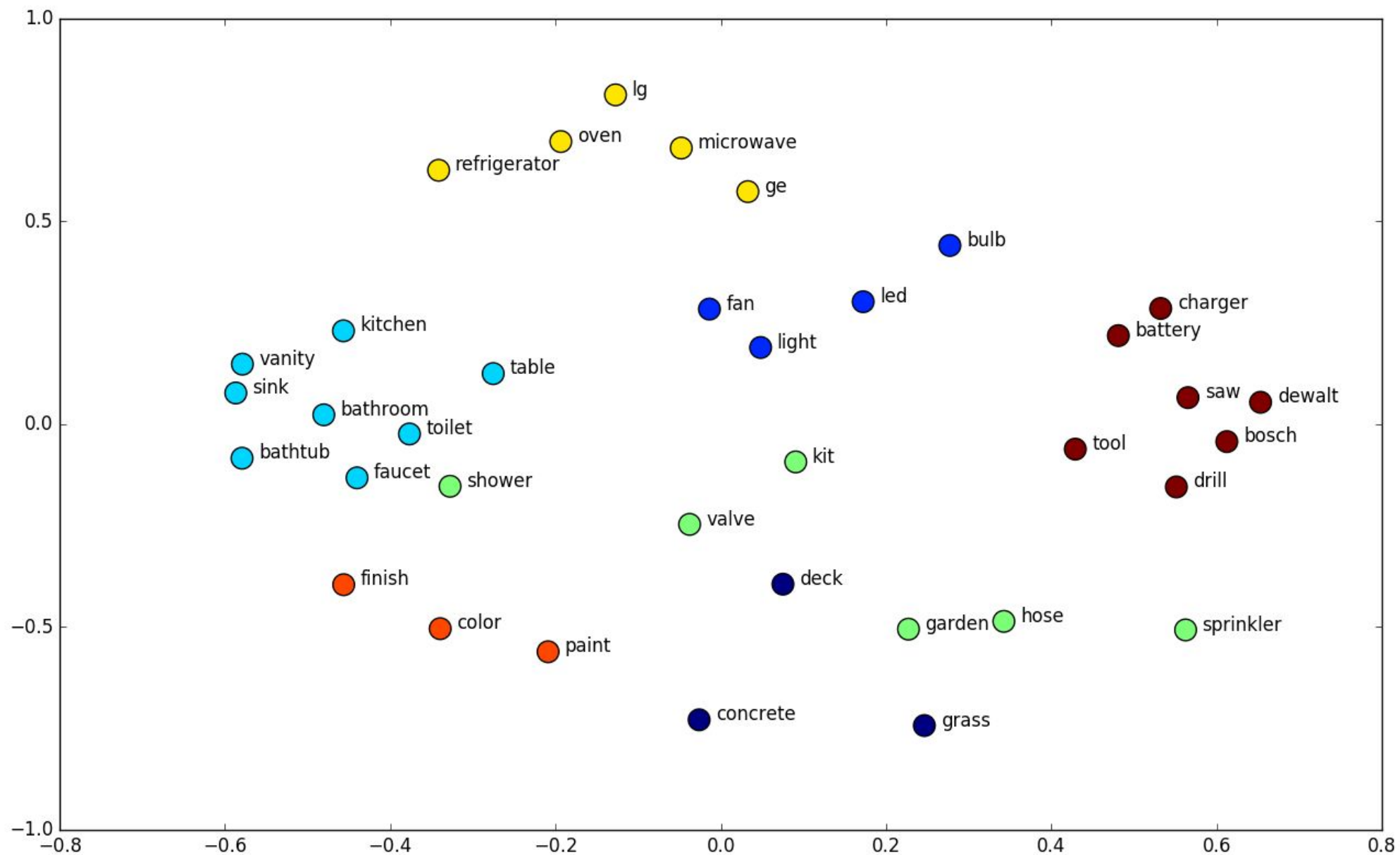
Points in space

Osgood et al. (1957) noticed that in using these 3 numbers to represent the meaning of a word, the model was representing each word as a point in a three-dimensional space

Part of the meaning of *heartbreak* can be represented as a vector with three dimensions corresponded to the word's rating on the three scales.

heartbreak	2.45	5.65	3.58
------------	------	------	------

Vector Space Models



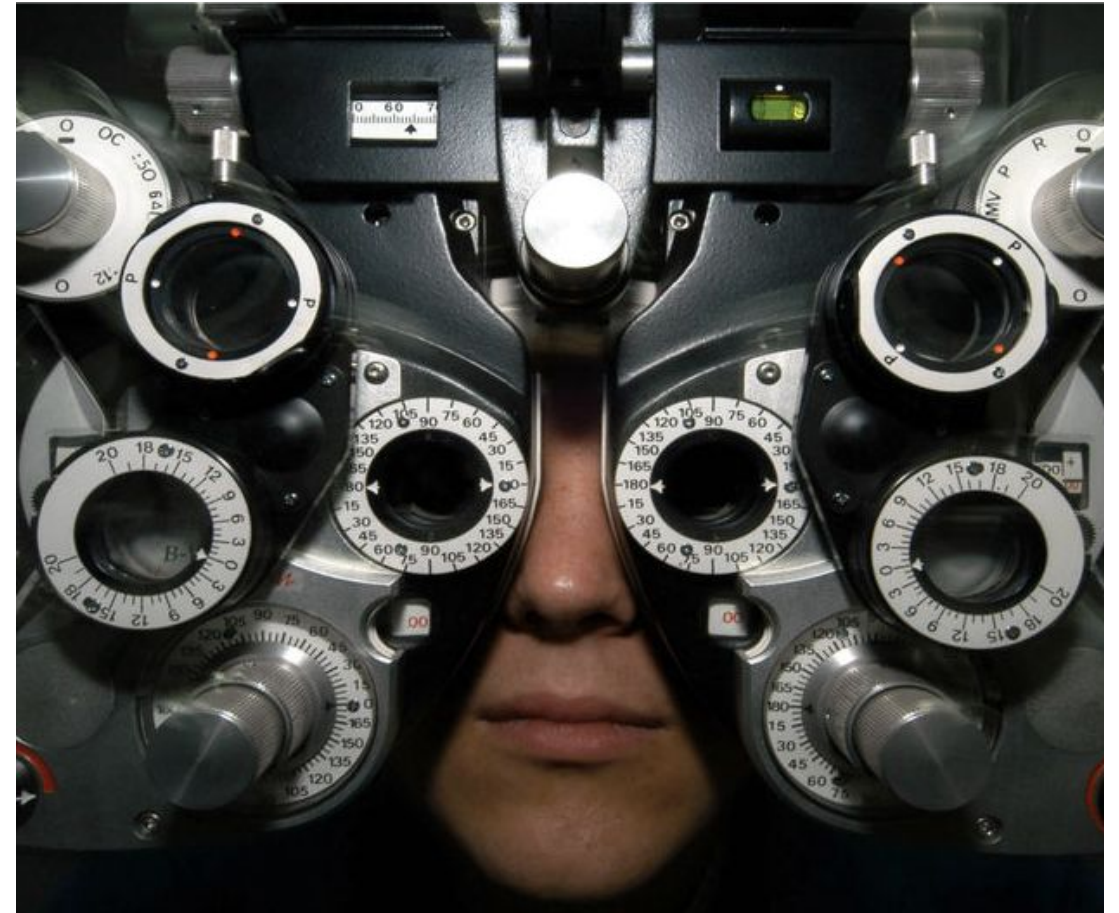
Distributional Hypothesis

If we consider *optometrist* and *eye-doctor* we find that, as our corpus of utterances grows, these two occur in almost the same environments. In contrast, there are many sentence environments in which *optometrist* occurs but *lawyer* does not...

It is a question of the relative frequency of such environments, and of what we will obtain if we ask an informant to substitute any word he wishes for optometrist (not asking what words have the same meaning).

These and similar tests all measure the probability of particular environments occurring with particular elements... If A and B have almost identical environments we say that they are synonyms.

–Zellig Harris (1954)



Intuition of distributional word similarity

Nida (1975) example:

A bottle of **tesgüino** is on the table
Everybody likes **tesgüino**
Tesgüino makes you drunk
We make **tesgüino** out of corn.

From context words humans can guess **tesgüino** means
an alcoholic beverage like beer

Intuition for algorithm:

Two words are similar if they have similar word contexts.

History of Vector Space Models

Vector Space Models were initially developed in the SMART information retrieval system (Salton, 1971)

Each document in a collection is represented as point in a space (a vector in a vector space)

A user's query is a pseudo-document and is represented as a point in the same space as the documents

Perform IR by retrieving documents whose vectors are close together in this space to the query vector

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

Each column vector represents a Document

Term-Document Matrix

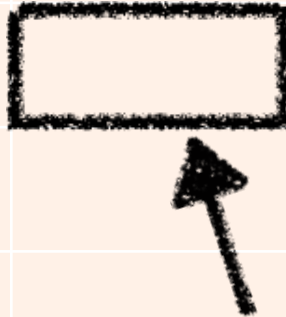
	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					



Each row vector
represents a
Term

Term-Document Matrix

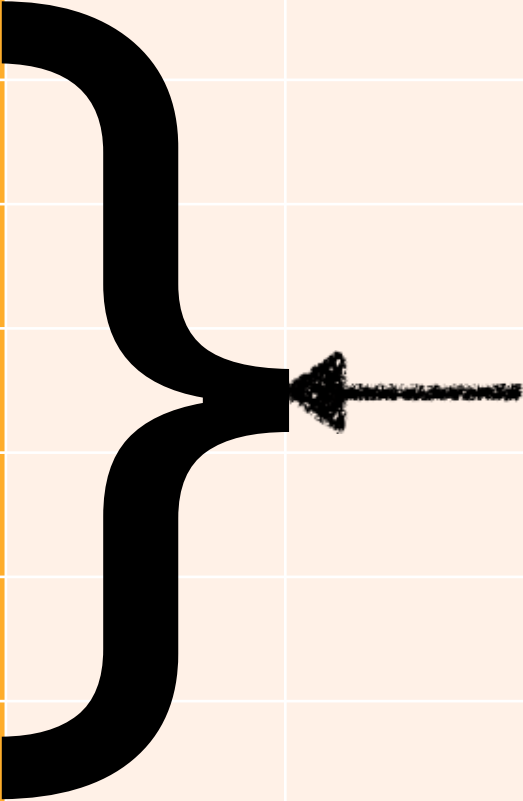
	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					



The value in a cell is
based on how often that term
occurred in that document

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					



The length of the document vectors is the size of the vocabulary

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

Document vectors
can be sparse
(most values are 0)

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

We can measure how similar two documents are by comparing their column vectors

What can document similarity let you do?

Word similarity for plagiarism detection

MAINFRAMES

Mainframes **are primarily** referred to large computers with **rapid**, advanced processing capabilities that **can execute and** perform tasks **equivalent to many** Personal Computers (PCs) machines **networked together**. It is **characterized with high quantity** Random Access Memory (RAM), very large secondary storage devices, and **high-speed** processors to cater for the needs of the computers under its service.

Consisting of advanced components, mainframes have the capability of running multiple large applications required by **many and** most enterprises **and organizations**. **This is** one of its advantages. Mainframes are also suitable to cater for those applications **(programs)** or files that are of very **high** demand by its users (clients). Examples of **such organizations and enterprises using mainframes are** online shopping websites **such as** Ebay, Amazon, **and computing-giant**

MAINFRAMES

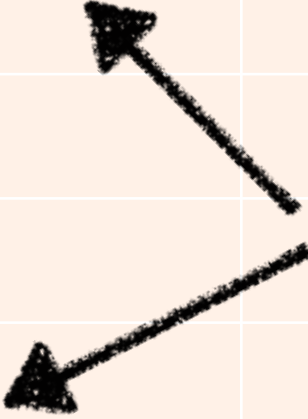
Mainframes **usually are** referred those computers with **fast**, advanced processing capabilities that **could perform by itself** tasks **that may require a lot of** Personal Computers (PC) Machines. **Usually mainframes would have lots of** RAMs, very large secondary storage devices, and **very fast** processors to cater for the needs of those computers under its service.

Due to the advanced components mainframes have, **these computers** have the capability of running multiple large applications required by most enterprises, **which is** one of its advantage. Mainframes are also suitable to cater for those applications or files that are of very **large** demand by its users (clients). Examples of these **include** the large online shopping websites **-i.e. : Ebay, Amazon, Microsoft, etc.**

Term-Document Matrix

	D1	D2	D3	D4	D5
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

What does comparing two row vectors do?



Vector comparisons

	doc_x	doc_y
A	2	4
B	10	15
C	14	10

Vector comparisons

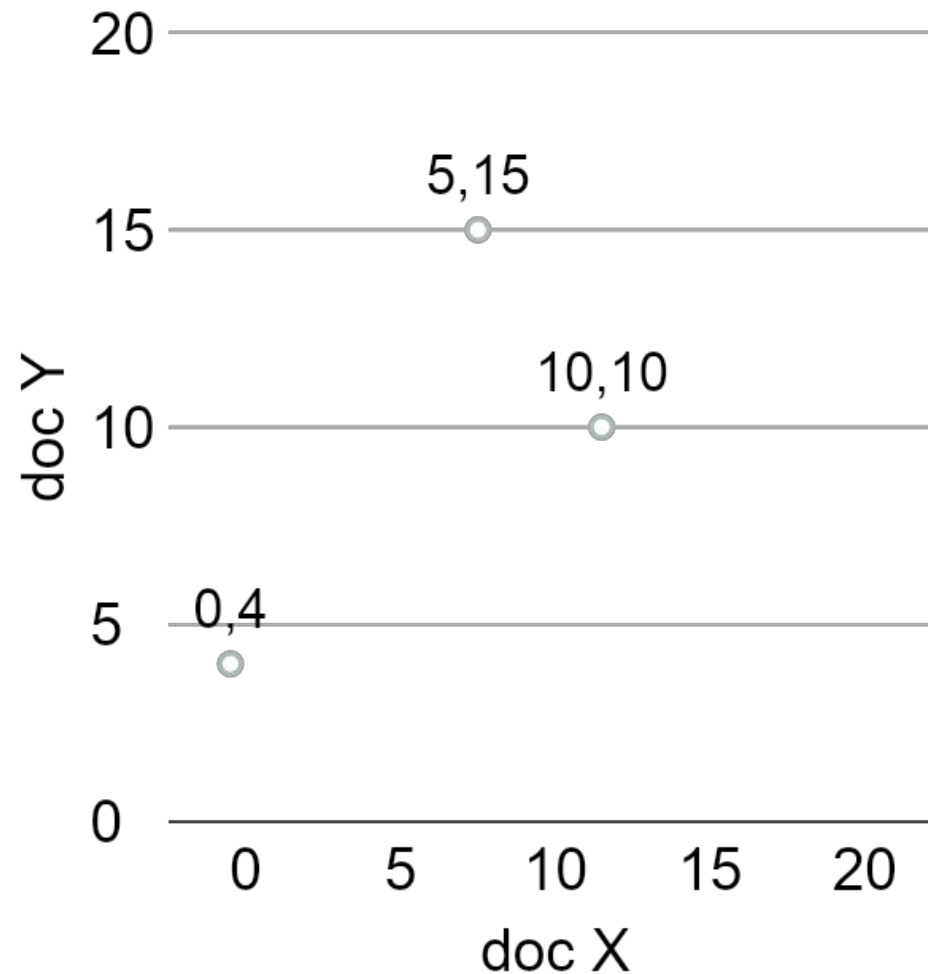
	doc_x	doc_y
A	2	4
B	10	15
C	14	10

doc_y is a positive movie review
 doc_x is a less positive movie review

A = "superb" positive / low frequency
B = "good" positive / high frequency
C = "disappointing" negative / high frequency

Vector comparisons

	doc_x	doc_y
A	2	4
B	10	15
C	14	10

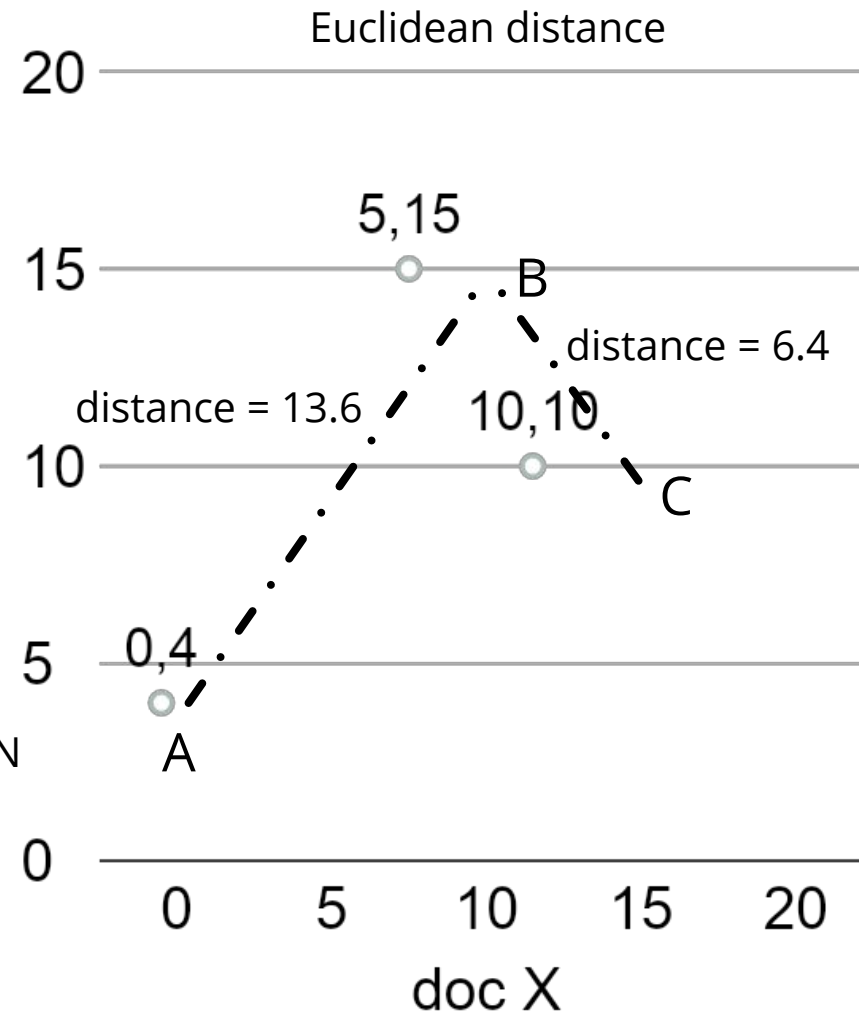


Vector comparisons

	doc_x	doc_y
A	2	4
B	10	15
C	14	10

Euclidean distance : vectors u, v of dimension N

$$\sqrt{\sum_{i=1}^N |u_i - v_i|^2}$$



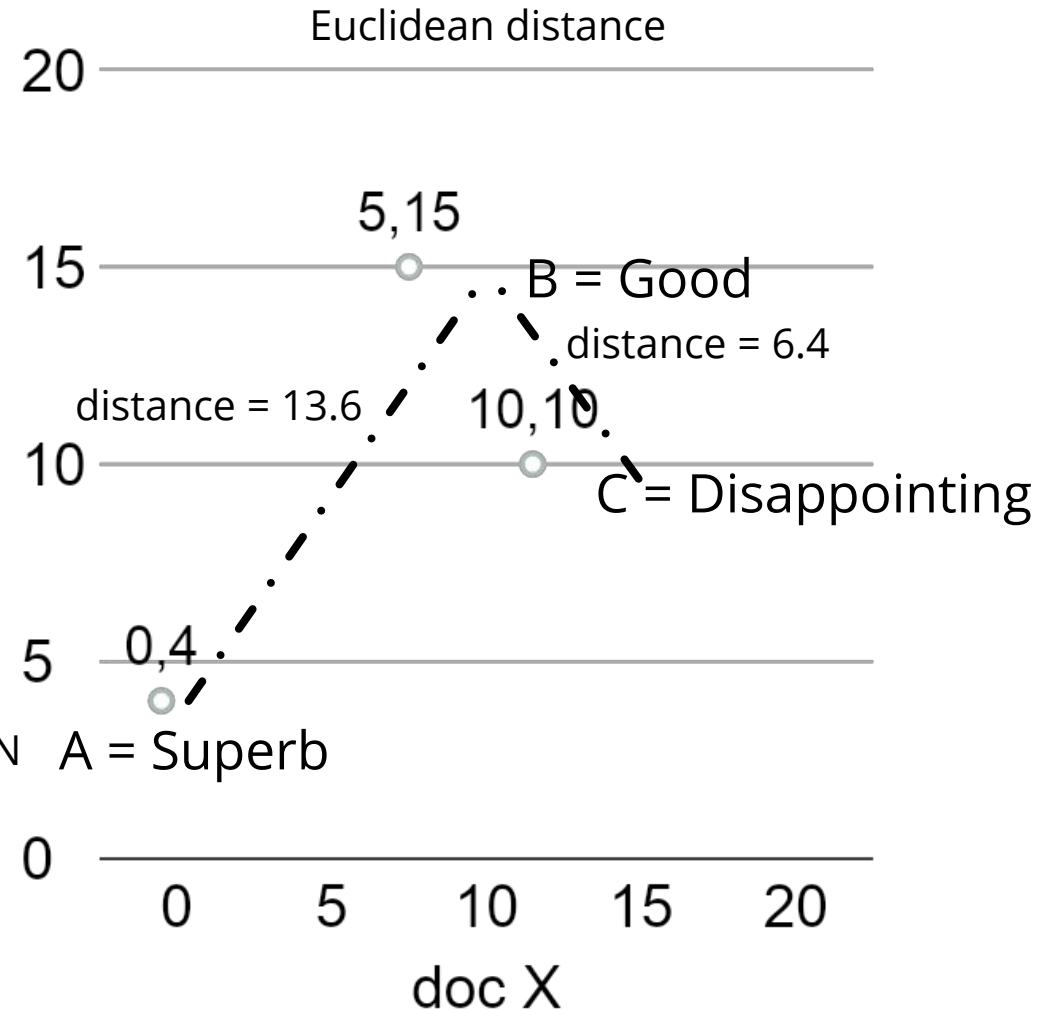
Vector comparisons

	doc_x	doc_y
A	2	4
B	10	15
C	14	10

Euclidean distance : vectors u, v of dimension N

$$\sqrt{\sum_{i=1}^N |u_i - v_i|^2}$$

Oh no! Good is closer to Disappointing than to Superb.



Vector L2 (length) Normalization

	doc_x	doc_y	$\ u\ $
A	2	4	4.47
B	10	15	18.02
C	14	10	17.20

$$\|u\| = \sqrt{\sum_{i=1}^n u_i^2}$$

Vector L2 (length) Normalization

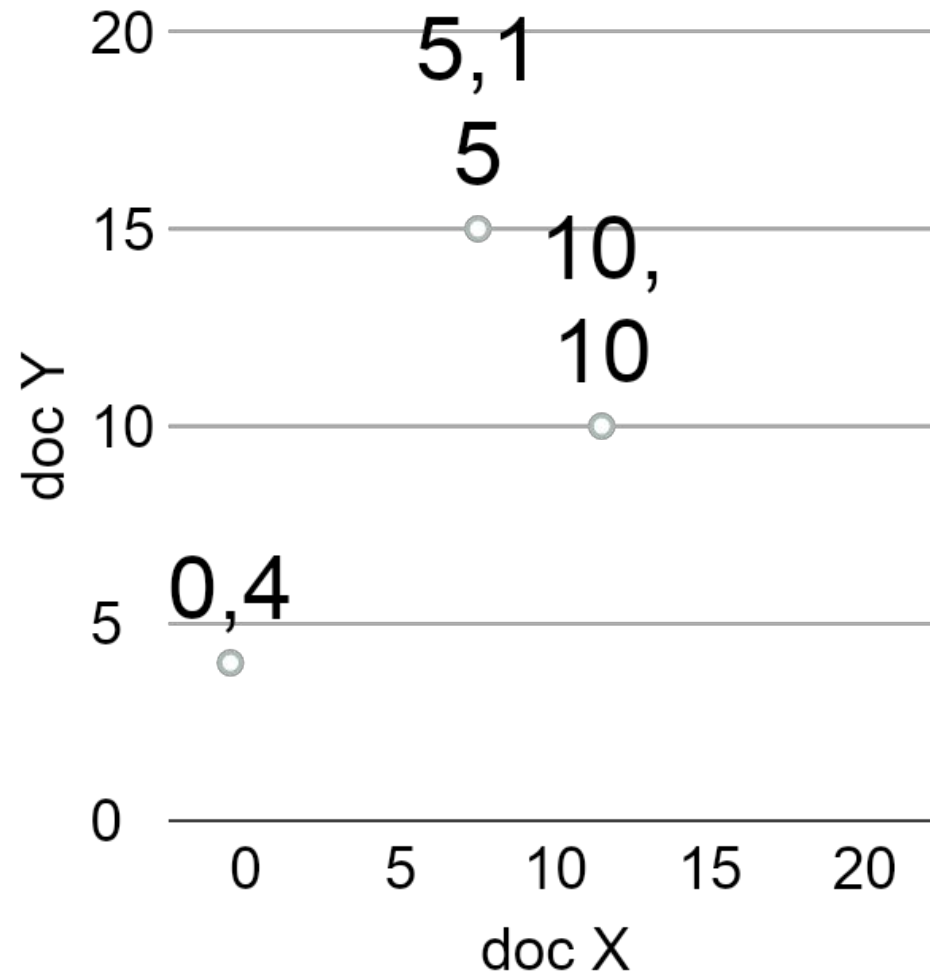
	doc_x	doc_y	$\ u\ $
A	2/4.47	4/4.47	4.47
B	10/18.02	15/18.02	18.02
C	14/17.2	10/17.2	17.20

$$\|u\| = \sqrt{\sum_{i=1}^n u_i^2}$$

Divide each vector by its L2 length

Vector comparisons

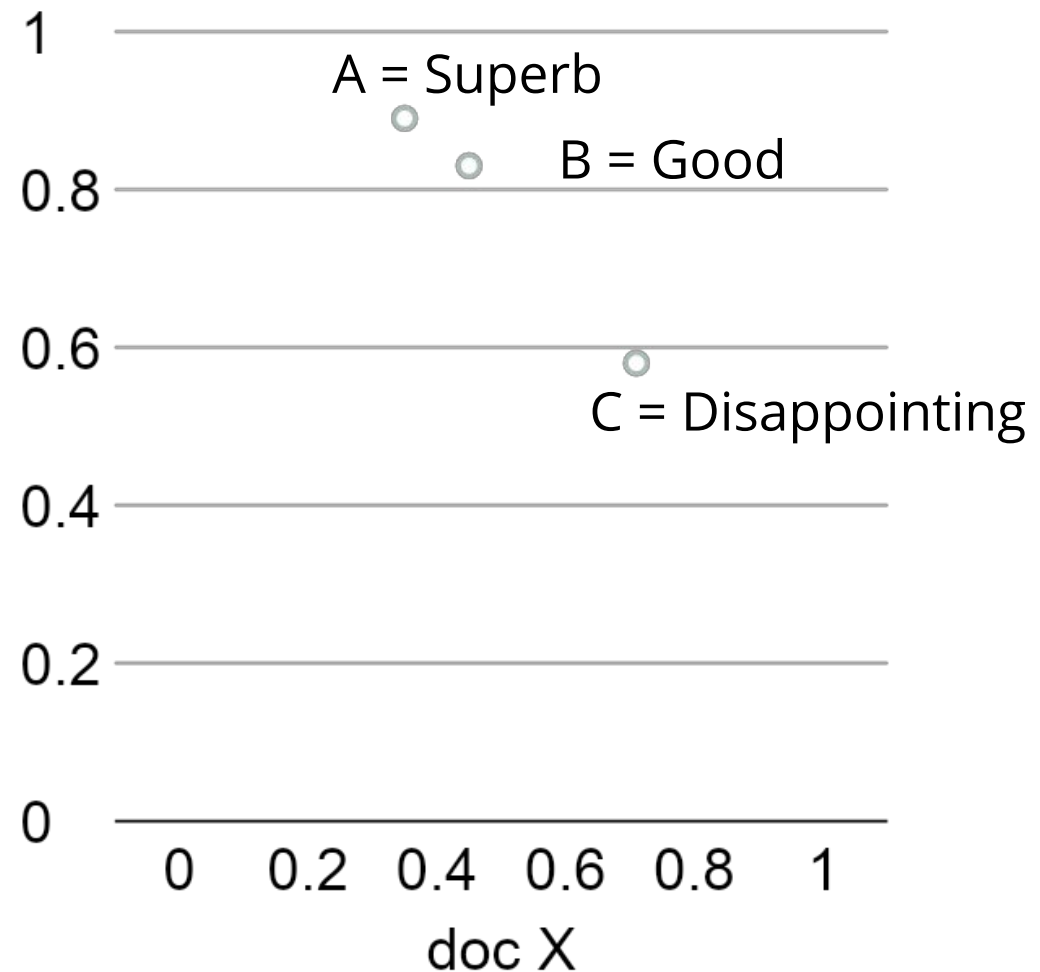
doc_X	doc_Y
2	4
10	15
14	10



Vector L2 (length) Normalization

	doc_x	doc_y
A	0.45	0.89
B	0.55	0.83
C	0.81	0.58

Now Good is
closer to Superb
than to
Disappointing



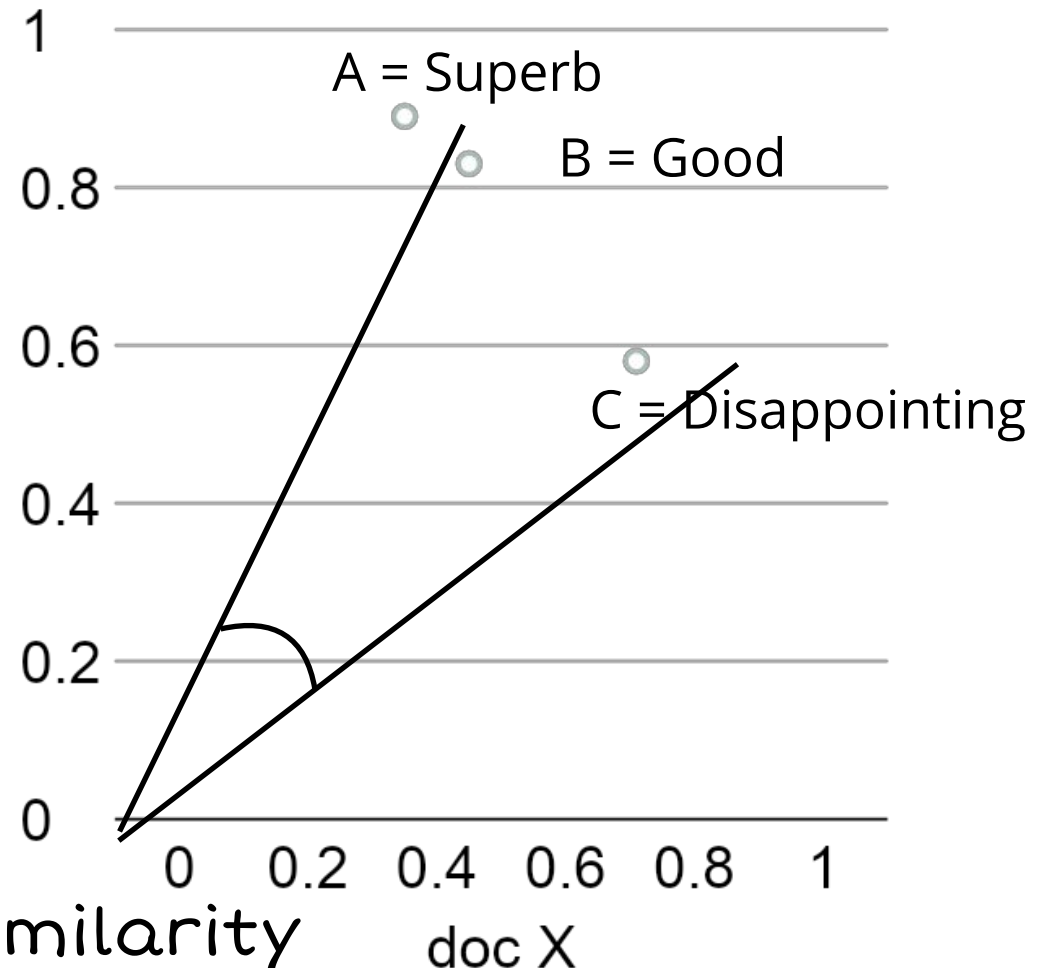
Cosine Distance

$$1 - \frac{\sum_{i=1}^n u_i \times v_i}{\sqrt{\sum_{i=1}^n u_i^2} \times \sqrt{\sum_{i=1}^n v_i^2}}$$



Cosine does the L2 normalization too

Cosine angle between vectors tells us their similarity



Term-Term Matrix

	abandon	abdicate	abhor	...	zymurgy
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

Term-Term Matrix

AKA
Term-Context
Matrix

	abandon	abdicate	abhor	...	zymurgy
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

Length of the vector is now $|V|$
instead of number of documents



Term-Term Matrix

AKA
Term-Context
Matrix

	abandon	abdicate	abhor	...	zymurgy
abandon					
abdicate					
abhor					
academic					
...					
zygodactyl					
zymurgy					

The value in a cell indicates how often abandon appears in a context window surrounding abdicate

Context windows

$w-2, w-1$ **target_word** $w+1 w+2$

The government most not **abdicate** responsibility to non-elected
it has led men to **abdicate** their family responsibilities
other demands, but declining to **abdicate** his responsibility
leaders **abdicate** their role and present people with no plans

	his	leaders	not	responsibilit y	to
abdicate	1	1	1	2	3

Context windows

Occur in a window of +/- 2 words, in the same sentence, in the same document

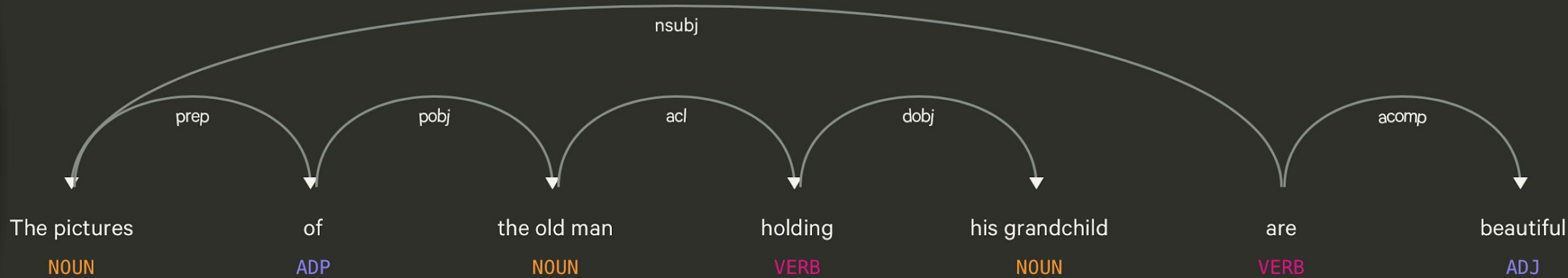
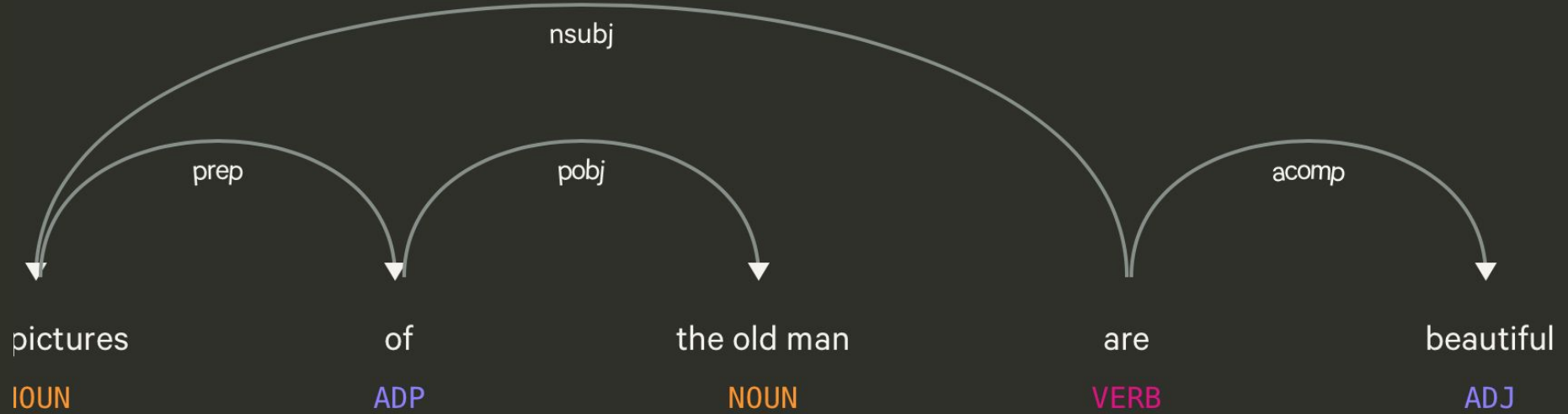
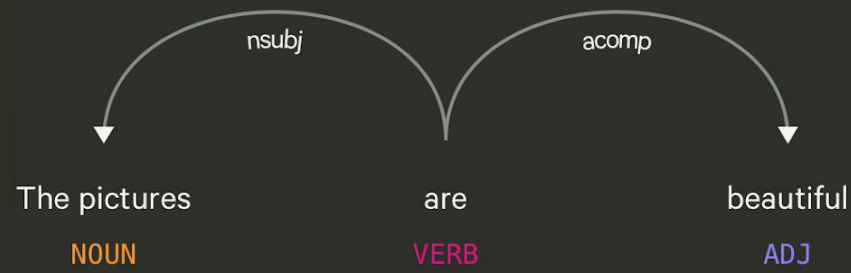
Instead of window of words use more complex contexts: dependency patters. Subj-of-verb, adj-mod, obj-of-verb

Languages have long distance dependencies

*The **pictures are** beautiful.*

*The **pictures of the old man are** beautiful.*

*The **pictures of the old man holding his grandchild are** beautiful.*



Using syntax to define a word's context

Zellig Harris (1968) "The meaning of entities, and the meaning of grammatical relations among them, is related to the restriction of combinations of these entities relative to other entities"

Duty and Responsibility have similar syntactic distributions

Modified by adjectives

additional, administrative, assumed,
collective, congressional, constitutional
...

Object of verbs

assert, assign, assume, attend to, avoid,
become, breach..

Alternates to counts

Raw word frequency is not a great measure of association between words. It's very skewed "the" and "of" are very frequent, but maybe not the most discriminative

We'd rather have a measure that asks whether a context word is particularly informative about the target word.

Instead of raw counts, it's common to transform vectors using TF-IDF or PPMI

TF-IDF

*Term frequency * inverse document frequency*



How often a
word occurred in
a document



1 over the number
of documents that it
occurred in

Sparse v. Dense Vectors

Co-occurrence matrix (weighted by TF-IDF or mutual information)

- **Long** (length $|V| = 50,000+$)
- **Sparse** (most elements are zeros)

Alternative: learn vectors that are

- **Short** (length 200-1000)
- **Dense** (most elements are non-zero)

How do we get dense vectors?

One recipe: train a classifier!

1. Treat the target word and a neighboring context word as positive examples.
2. Randomly sample other words in the lexicon to get negative samples.
3. Use logistic regression to train a classifier to distinguish those two cases.
4. Use the weights as the embeddings.

Word2Vec

Mikolov et al. 2013

Learn embeddings as part of the process of word prediction.

Train a classifier to predict neighboring words

Inspired by neural net language models.

In so doing, learn dense embeddings for the words in the training corpus.

Advantages:

Fast, easy to train (much faster than SVD)

Available online in the word2vec package
Including sets of pretrained embeddings!

Word2Vec

Predict each neighboring word in a context window of $2C$ of surrounding words

So for $C=2$, we are given a word w_t and we try to predict its 4 surrounding words

$$[w_{t-2}, w_{t-1}, w_{t+1}, w_{t+2}]$$

Uses "negative sampling" for training

Negative sampling

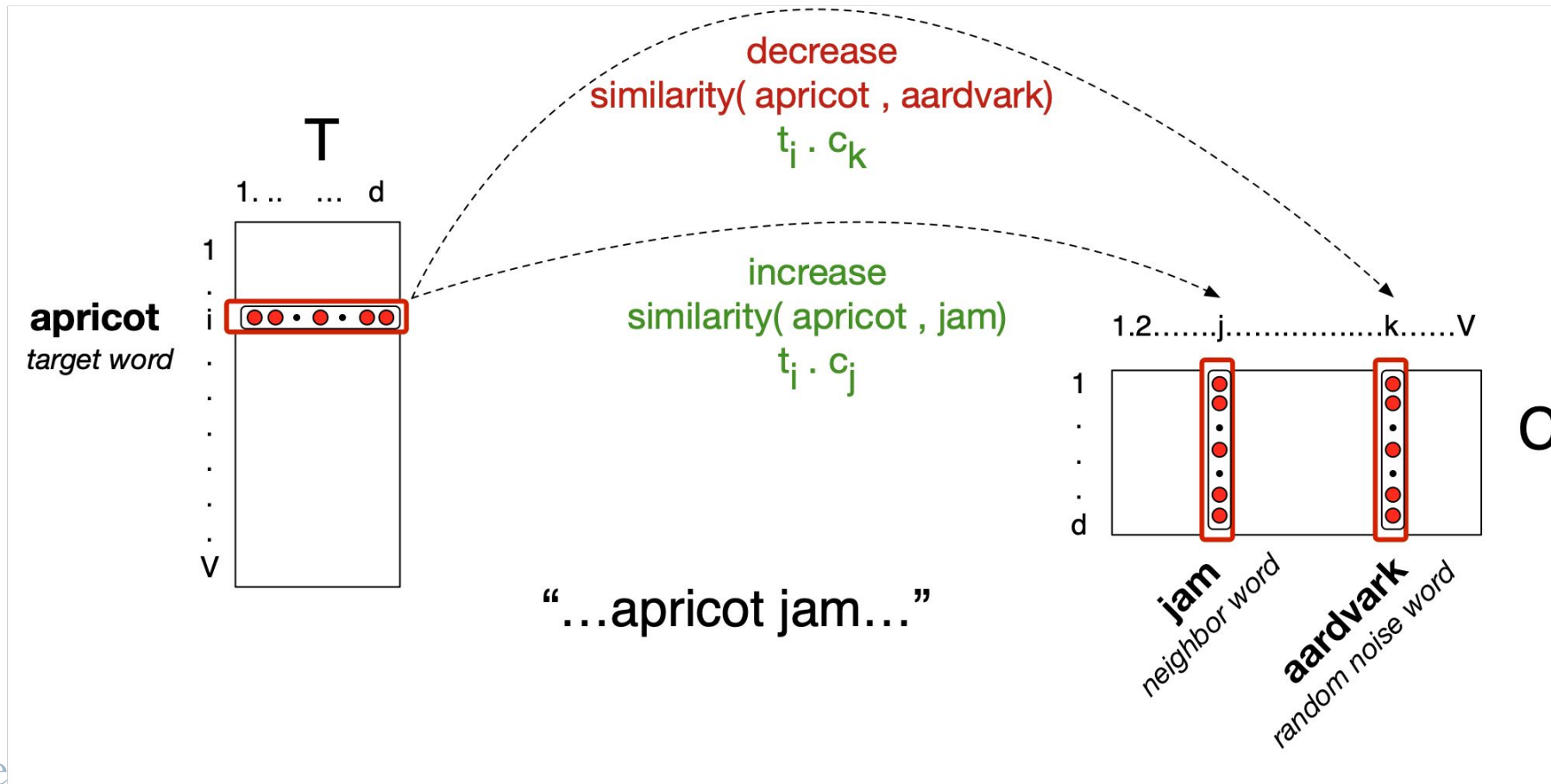
lemon, a [tablespoon of apricot preserves or] jam
c1 c2 w c3 c4

We want predictions
of these words to be high

And these words to be low

[cement metaphysical dear coaxial apricot attendant whence forever puddle]
n1 n2 n3 n4 n5 n6 n7 n8

Neural Network



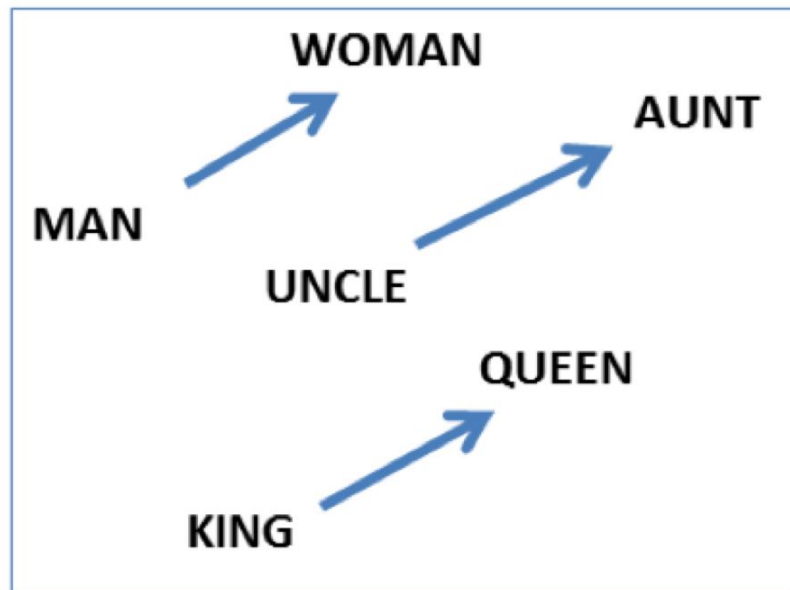
Properties of Embeddings

Nearest Neighbors are surprisingly good

target:	Redmond	Havel	ninjutsu	graffiti	capitulate
	Redmond Wash.	Vaclav Havel	ninja	spray paint	capitulation
	Redmond Washington	president Vaclav Havel	martial arts	grafitti	capitulated
	Microsoft	Velvet Revolution	swordsmanship	taggers	capitulating

Embeddings capture relational meanings

$\text{vector('king')} - \text{vector('man')} + \text{vector('queen')} \approx \text{vector('woman')}$



Magnitude: A Fast, Efficient Universal Vector Embedding Utility Package

Ajay Patel

Plasticity Inc.

San Francisco, CA

ajay@plasticity.ai

Alexander Sands

Plasticity Inc.

San Francisco, CA

alex@plasticity.ai

Chris Callison-Burch

Computer and Information

Science Department

University of Pennsylvania

ccb@upenn.edu

Marianna Apidianaki

LIMSI, CNRS

Université Paris-Saclay

91403 Orsay, France

marapi@seas.upenn.edu

Abstract

Vector space embedding models like word2vec, GloVe, and fastText are extremely popular representations in natural language processing (NLP) applications. We present Magnitude, a fast, lightweight tool for utilizing and processing embeddings. Magnitude is an open source Python package with a compact vector storage file format that allows for efficient manipulation of huge numbers of embeddings. Magnitude performs common operations up to 60 to 6,000 times faster than Gensim. Magnitude introduces several novel features for improved robustness like

Metric	Cold	Warm
Initial load time	97x	–
Single key query	1x	110x
Multiple key query (n=25)	68x	3x
k-NN search query (k=10)	1x	5,935x

Table 1: Speed comparison of Magnitude versus Gensim for common operations. The ‘cold’ column represents the first time the operation is called. The ‘warm’ column indicates a subsequent call with the same keys.

file, a 97x speed-up. Gensim uses 5GB of RAM versus 18KB for Magnitude.

Demo of word vectors

Install Magnitude

```
pip3 install pymagnitude
```

Download Google's word2vec vectors

```
wget
```

```
http://magnitude.plasticity.ai/word2vec+approx/GoogleNews-vectors-negative300.magnitude
```

Warning it's 11GB large

Start Python, and try the commands

on the next slide

```
python3
```

Demo of word vectors

```
from pymagnitude import *  
vectors = Magnitude("GoogleNews-vectors-negative300.magnitude")
```

```
queen = vectors.query('queen')  
king = vectors.query("king")  
vectors.similarity(king, queen)  
# 0.6510958
```

```
vectors.most_similar_approx(king, topn=5)  
#[('king', 1.0), ('kings', 0.72), ('prince', 0.62), ('sultan', 0.59), ('ruler',  
0.58)]
```


Many possible models

Matrix type	Reweighting	Comparisons
Term-document	length norm.	cosine
Term-context	TF-IDF	Manhattan
Pattern-pair	PPMI	Jaccard
Dim. Reduction	probabilities	KL divergence
word2vec	How many dimensions?	JS distance
GloVe		DICE
PCA	What modifications should we make to the input?	
LDA		
LSA		